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1 ###
2 import numpy as np
3 import pandas as pd
4 import warnings
5 warnings.filterwarnings("ignore")
6 ### md
7 #### Importing the dataset comprising data of
  children with illegal immigration status or no parent
  or guardian
8 ###
9 df = pd.read_csv("
  HHS_Unaccompanied_Alien_Children_Program.csv")
10 df.head()
11 ### md
12 #### Checking for null values
13 ###
14 df.isna().sum()
15 ### md
16 #### Converting data to pandas date-time format
17 ###
18 df["Date"] = pd.to_datetime(df["Date"])
19 df.head()
20 ### md
21 #### Arrange dates in ascending order
22 ###
23 df.sort_values(by=["Date"], inplace=True)
24 df.reset_index(drop=True, inplace=True)
25 df.head()
26 ### md
27 #### Retrieving weekday names for each date in the
  dataset
28 ###
29 df["Day of Week"] = list(map(lambda x: x.strftime("%A
  "), df["Date"]))
30 df = df[["Date", "Day of Week", "Children apprehended
  and placed in CBP custody*", "Children in CBP custody"
  , "Children transferred out of CBP custody", "Children
  in HHS Care", "Children discharged from HHS Care"]]
31 df.head()
32 ### md
33 #### Finding out the total number of records on each
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33 day of week
34 #%%
35 weekday_number = df["Day of Week"].value_counts().
  reset_index()
36 weekday_number["Day of Week Number"] = weekday_number
  ["Day of Week"].map({
37     "Sunday": "1",
38     "Monday": "2",
39     "Tuesday": "3",
40     "Wednesday": "4",
41     "Thursday": "5",
42     "Friday": "6",
43     "Saturday": "7"
44 })
45 weekday_number = weekday_number.sort_values(by=["Day
  of Week Number"]).reset_index(drop=True)
46 weekday_number
47 #%% md
48 Looking at the numbers above, we can see that the
  number of days on which children were recorded is
  relatively consistent from Sunday through Thursday.
  From Monday through Thursday, the figures range
  between 145 and 150 days. Sunday is somewhat lower,
  at 130 days, but it is still reasonably close to this
  range.
49
50 Friday, however, stands out as an exceptionally low
  number of recorded days. Data is recorded for only
  two Fridays, compared with more than 130 recorded
  days for each of the other weekdays. This substantial
  difference suggests that Friday may represent a
  distinct pattern in the data and warrants further
  investigation.
51
52 There is one assumption that we can make here, which
  is deleting those two Friday records and assuming
  that services are not provided on Fridays and
  Saturdays. But in reality, these services are
  provided 24 hours a day and 7 days a week. So we need
  to include these days on which data is not recorded
  and then somehow deal with the null values for number

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52 of children on these days.
53 ### md
54 #### Creating a new dataset with dates on which data
    is not recorded
55 ###
56 df_2 = pd.DataFrame(columns=df.columns)
57 df_2["Date"] = np.setdiff1d(pd.date_range(df["Date"].
    min(),df["Date"].max()).values,df["Date"].values)
58 df_2["Day of Week"] = list(map(lambda x: x.strftime(
    "%A"), df_2["Date"]))
59 df_2[['Children apprehended and placed in CBP custody
    *','Children in CBP custody','Children transferred
    out of CBP custody','Children in HHS Care', 'Children
    discharged from HHS Care']] = np.nan
60 df_2.head()
61 ### md
62 #### Concatenating the original dataset with the new
    dataset comprising unrecorded dates
63 ###
64 df_3 = pd.concat([df,df_2]).sort_values(by=["Date"]).
    reset_index(drop=True)
65 df_3.drop(columns=["Day of Week"], inplace=True)
66 df_3.set_index("Date", inplace=True)
67 df_3.head()
68 ### md
69 #### Checking the data types of each feature
70 ###
71 df_3.dtypes
72 ### md
73 As we observe the above data types, the feature *
    Children in HHS care* is of object type. The other
    features are of float type. In reality, these
    features denote number of children going through
    their respective stage. This means that such features
    ought to be integers.
74
75 All the features except the *Children in HHS Care*
    are floating points, which makes it unnecessary for
    us to convert into integers, as they are of numeric
    type. The values for such features end with ".0". The
    *Children in HHS Care* is of object type, which is a

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75 non-numeric type - this forces us to convert it to numeric type. Before type conversion, let us see how the values of *Children in HHS Care* look like:

76 #%%

77 print(df_3["Children in HHS Care"].unique().tolist())

78 #%% md

79 As we observe, there are commas in each element of *Children in HHS Care* column. We need to remove those commas and convert the data type into floating point so that the data type matches with that of other columns.

80 #%%

81 df_4 = df_3.copy()

82

83 df_4["Children in HHS Care"] = list(map(lambda x: "".join(x.split(",")), df_4["Children in HHS Care"]))

84 #%% md

85 From the above error, we can say that there are some floating point values for the column *Children in HHS Care*. We need to explicitly convert all values to string, then remove those commas and then convert them to floating point.

86 #%%

87 df_4["Children in HHS Care"] = list(map(lambda x: "".join(str(x).split(",")), df_4["Children in HHS Care"]))

88 df_4["Children in HHS Care"] = df_4["Children in HHS Care"].astype("float64")

89 df_4.head()

90 #%% md

91 Now, let us make use of data interpolation, which figures out values for each null value by itself based on non-null values around each of them.

92 #%%

93 df_4 = df_4.interpolate().copy()

94 df_4.head()

95 #%% md

96 As we observe the above dataset, the values are in decimal points. In reality the features denote number of children in each phase - this means that

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96 the elements should also be integers. So, we shall
   round all the elements to their nearest integers.
97 #%%
98 df_5 = df_4.round(0)
99 df_5.head()
100 #%% md
101 Now, we need to have a look for patterns for each
   column. The best way is to visualise the data.
102 #%%
103 import matplotlib.pyplot as plt
104 import seaborn as sns
105 #%%
106 df_6 = df_5.reset_index()
107 df_6.head()
108 #%% md
109 Let us export the data to a CSV file. We shall
   perform data visualisation in Tableau.
110 #%%
111 df_6.to_csv("HHS_Program_Cleaned.csv")
112 #%% md
113 ![image.png](attachment:8349c1fb-5268-44f9-b861-
   57302017ca4f.png)
114 #%% md
115 ![image.png](attachment:75c8d246-b484-45fe-9503-
   9eff84cc9989.png)
116 #%% md
117 ![image.png](attachment:b02de321-980f-4c2b-a380-
   df84bbf275ac.png)
118 #%% md
119 ![image.png](attachment:532435be-a0f1-4e5b-af60-
   80611a4e5d9c.png)
120 #%% md
121 From the above graphs, we can infer that Children
   who were considered by CBP, those already going
   through CBP custody and those done with the CBP
   custody and transferred to HHS care follow a very
   similar pattern. Same is the case with the children
   in HHS care and those who got discharged from it.
122
123 There is another important aspect extracted from the
   dataset. The number of children undergoing each

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123 phase of the program has gone drastically down in February 2025, and the same level is maintained throughout the rest of the time. This is mainly due to decrease in illegal immigration to the United States, thanks to harsh restrictions set up by the Trump government. Trump came into power on January 20, 2025. Some of his initial executive orders include crackdown of illegal immigration - this is the main driver behind numbers falling down.

124 #%%

125